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Question 1: Mapping Data onto Aesthetics

Scenario: You have a dataset containing information about product sales. You want to understand the relationship between Sales, Profit, and Category.

Dataset: Product Sales (₹ Lakhs), Profit (%), Category (Electronics, Clothing, Furniture).

Tasks:

1. Create the dataset for Question 1.
2. Write an R program to create a scatter plot with Sales on the x-axis and Profit on the y-axis. Map Category to color and Sales to size.
3. Interpret the relationship between Sales and Profit from the plot.
4. Identify the variable scales for Sales, Profit, and Category.
5. Suggest an alternative visualization for this data and explain why it might be useful.
6. (Optional) Provide R code for the suggested alternative visualization.

Answer:

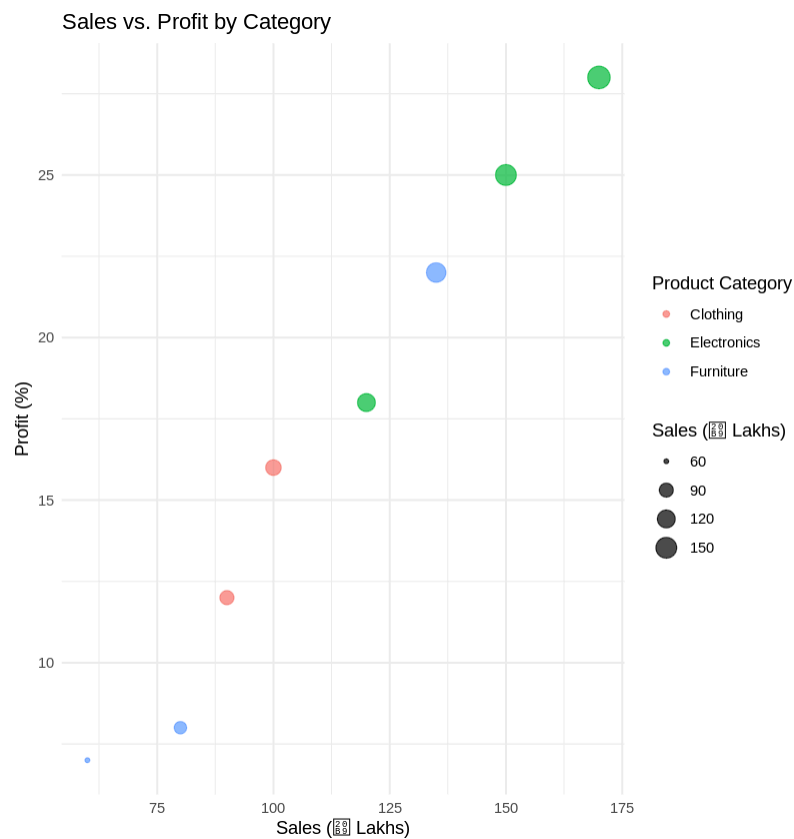
```
if (!requireNamespace("ggplot2", quietly = TRUE)) {  
  install.packages("ggplot2")  
}  
library(ggplot2)  
data_q1 <- data.frame(  
  Product = c("P1", "P2", "P3", "P4", "P5", "P6", "P7", "P8"),  
  Sales = c(120, 90, 150, 80, 135, 100, 170, 60), # in Lakhs  
  Profit = c(18, 12, 25, 8, 22, 16, 28, 7), # in Percent  
  Category = c("Electronics", "Clothing", "Electronics", "Furniture", "Furniture", "Clothing", "Electronics", "Furniture")  
)  
  
print("Dataset for Question 1:")  
print(data_q1)  
scatter_plot_q1 <- ggplot(data_q1, aes(x = Sales, y = Profit, color = Category, size = Sales)) +  
  geom_point(alpha = 0.7) +  
  labs(  
    title = "Sales vs. Profit by Category",  
    x = "Sales (₹ Lakhs)",  
    y = "Profit (%)",  
    color = "Product Category",  
    size = "Sales (₹ Lakhs)"  
  )
```

```
) +  
theme_minimal()
```

```
print(scatter_plot_q1)
```

[1] "Dataset for Question 1:"

	Product	Sales	Profit	Category
1	P1	120	18	Electronics
2	P2	90	12	Clothing
3	P3	150	25	Electronics
4	P4	80	8	Furniture
5	P5	135	22	Furniture
6	P6	100	16	Clothing
7	P7	170	28	Electronics
8	P8	60	7	Furniture



Interpretation for Question 1

3. Interpret the relationship between Sales and Profit from the plot.

From the scatter plot, a **generally positive correlation** can be observed between Sales and Profit. As Sales increase, Profit tends to increase as well.

- **Electronics** products (blue) tend to have higher sales and higher profit percentages compared to other categories.
- **Furniture** products (green) generally show lower sales and profit.

- **Clothing** products (red) fall in the middle range. The size of the points also visually reinforces that higher sales values correspond to larger points, making the high-sales, high-profit electronics products stand out.

4. Identify the variable scales for Sales, Profit, and Category.

- **Sales:** Ratio scale (numerical, has a true zero point, allows for ratios).
- **Profit:** Ratio scale (numerical, has a true zero point, allows for ratios).
- **Category:** Nominal scale (categorical, no inherent order).

5. Suggest an alternative visualization for this data and explain why it might be useful.

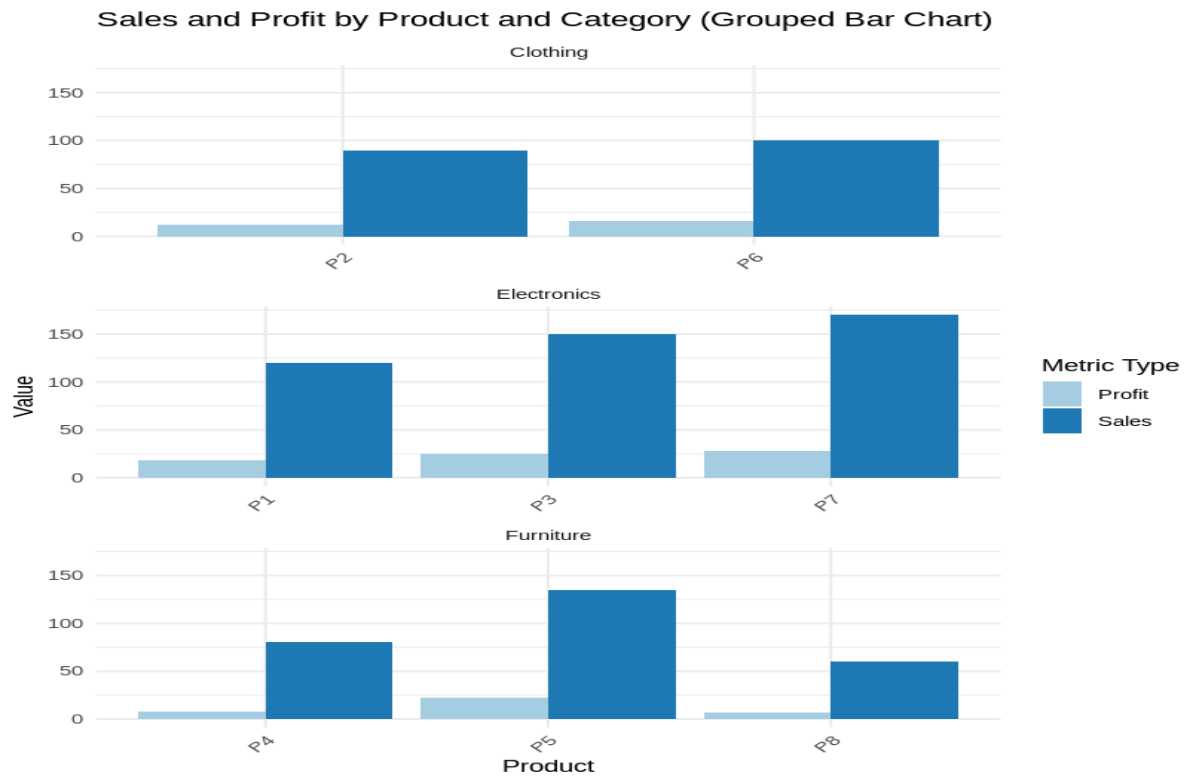
An alternative visualization could be a **Grouped Bar Chart**.

- **Why useful:** While the scatter plot shows the relationship between sales and profit, a grouped bar chart could be useful if the primary goal is to **directly compare the absolute Sales and Profit values across different Categories and potentially individual Products within those categories**. It provides a clearer side-by-side comparison of the two metrics.

```
# Install and load tidyr if not already installed
if (!requireNamespace("tidyr", quietly = TRUE)) {
  install.packages("tidyr")
}
library(tidyr)
data_q1_long <- tidyr::pivot_longer(data_q1, cols = c(Sales, Profit), names_to = "Metric", values_to = "Value")

grouped_bar_chart_q1 <- ggplot(data_q1_long, aes(x = Product, y = Value, fill = Metric)) +
  geom_bar(stat = "identity", position = "dodge") +
  facet_wrap(~ Category, scales = "free_x", ncol = 1) +
  labs(
    title = "Sales and Profit by Product and Category (Grouped Bar Chart)",
    x = "Product",
    y = "Value",
    fill = "Metric Type"
  ) +
  scale_fill_brewer(palette = "Paired") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

print(grouped_bar_chart_q1)
```



Question 2: Coordinate Systems (Cartesian and Polar)

Scenario: You have collected data on the frequency of different wind directions.

Dataset: Direction (N, NE, E, SE, S, SW, W, NW), Frequency (counts).

Tasks:

1. Create the dataset for Question 2.
2. Write an R program to visualize the dataset using a Cartesian coordinate system.
3. Write another R program to visualize the same data using a Polar coordinate system.
4. Compare the effectiveness of both visualizations.
5. Explain which coordinate system is more appropriate for directional data.
6. Justify your answer based on data interpretation.

```

if (!requireNamespace("ggplot2", quietly = TRUE)) {
  install.packages("ggplot2")
}
library(ggplot2)

directional_data_q2 <- data.frame(
  Direction = c("N", "NE", "E", "SE", "S", "SW", "W", "NW"),
  Frequency = c(18, 12, 20, 15, 22, 11, 16, 14)
)

print("Dataset for Question 2:")
print(directional_data_q2)

directional_data_q2$Direction <- factor(directional_data_q2$Direction,
                                         levels = c("N", "NE", "E", "SE", "S", "SW", "W", "NW"))

cartesian_plot_q2 <- ggplot(directional_data_q2, aes(x = Direction, y = Frequency, fill = Direction)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Directional Data in Cartesian Coordinates",
    x = "Direction",
    y = "Frequency"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
print(cartesian_plot_q2)

polar_plot_q2 <- ggplot(directional_data_q2, aes(x = Direction, y = Frequency, fill = Direction)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar(start = -pi/8) +
  labs(
    title = "Directional Data in Polar Coordinates (Wind Rose style)",
    x = "",
    y = "Frequency"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = seq(0, 360, by = 45), hjust = 0.5)) + # Adjust text angle
  theme(legend.position = "none")

print(polar_plot_q2)

```

"Dataset for Question 2:"

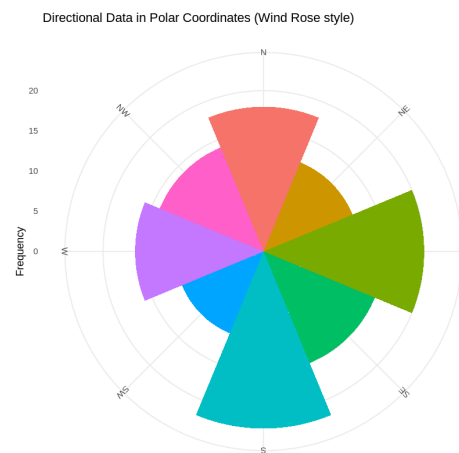
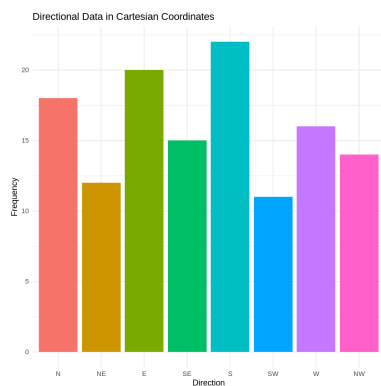
Direction Frequency

1	N	18
2	NE	12
3	E	20
4	SE	15
5	S	22
6	SW	11
7	W	16
8	NW	14

Warning message:

“Vectorized input to `element_text()` is not officially supported.

Results may be unexpected or may change in future versions of ggplot2.”



Interpretation for Question 2

4. Compare the effectiveness of both visualizations.

- **Cartesian Plot (Bar Chart):** This visualization clearly shows the frequency for each direction using bar heights. It's easy to compare the exact frequency values between different directions. However, it doesn't intuitively represent the circular nature or continuity of directional data. The discrete bars imply a separation that doesn't exist in the real-world phenomenon of direction.
- **Polar Plot (Wind Rose style):** This visualization uses a circular layout, which is highly effective for directional data. It visually emphasizes the circular relationship between directions and allows for an immediate perception of the distribution around a compass. Peaks and troughs in frequency are easily identifiable in their respective directional context. While comparing exact frequency magnitudes can be slightly more challenging than with a standard bar chart due to the radial axis, the overall pattern and cyclical nature are much clearer.

5. Explain which coordinate system is more appropriate for directional data.

The **Polar coordinate system** is generally more appropriate for directional data.

6. Justify your answer based on data interpretation.

Directional data, such as wind direction or compass bearings, inherently possesses a cyclical nature. For example, North follows Northwest, and North also follows Northeast. A Cartesian system (like a standard bar chart) treats each direction as a discrete, independent category, breaking this natural circularity. This can lead to a misinterpretation of the underlying patterns and relationships, as it fails to represent the continuity between directions.

The Polar coordinate system, by arranging the directions around a circle, visually respects this cyclical property. It allows the viewer to immediately grasp the overall distribution of frequencies across all directions and to see which directions are more prevalent and how they relate to adjacent directions. This intuitive representation is crucial for understanding spatial or cyclical phenomena, making it superior for interpreting directional data.

Question 3: Color Scales for Data Representation

Install and load ggplot2 if not already installed

```
if (!requireNamespace("ggplot2", quietly = TRUE)) {
```

```
  install.packages("ggplot2")
```

```
}
```

```
library(ggplot2)
```

```
student_marks <- data.frame(
```

```
  Student = c("A", "B", "C", "D", "E", "F"),
```

```
  Marks = c(45, 58, 72, 81, 67, 90)
```

```
)
```

```
print("Dataset for Question 3:")
```

```
print(student_marks)
```

Order students by marks for better visualization

```
student_marks$Student <- factor(student_marks$Student, levels =
```

```
student_marks$Student[order(student_marks$Marks)])
```

2. Write an R program to create a bar chart using a sequential color scale based on Marks.

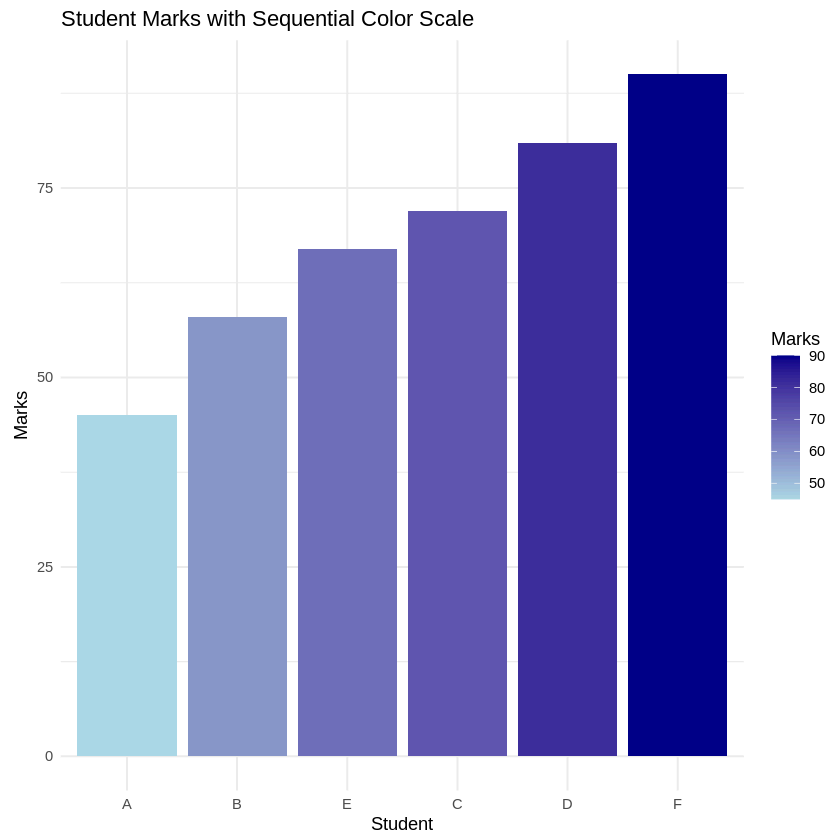
```
bar_chart_sequential_color <- ggplot(student_marks, aes(x = Student, y = Marks, fill =  
Marks)) +  
  geom_bar(stat = "identity") +  
  scale_fill_gradient(low = "lightblue", high = "darkblue") + # Sequential color scale  
  labs(  
    title = "Student Marks with Sequential Color Scale",  
    x = "Student",  
    y = "Marks",  
    fill = "Marks"  
  ) +  
  theme_minimal()
```

```
print(bar_chart_sequential_color)
```

"Dataset for Question 3:"

Student Marks

1	A	45
2	B	58
3	C	72
4	D	81
5	E	67
6	F	90



Interpretation for Question 3

3. Identify the students with the highest and lowest marks from the visualization.

- **Highest Marks:** Student F (90 Marks)
- **Lowest Marks:** Student A (45 Marks)

4. Explain why a sequential color scale is appropriate.

A sequential color scale is appropriate because the 'Marks' variable is quantitative and represents a continuous range of values from low to high. A sequential scale uses variations in lightness and/or saturation of a single hue, or a progression of hues, to effectively convey the order and magnitude of these numerical values. It allows for quick visual identification of lower (lighter colors) and higher (darker colors) mark ranges, making it easy to see performance differences at a glance.

5. Suggest a color palette suitable for color-blind users.

For color-blind users, it's best to use color palettes that rely heavily on differences in lightness and saturation rather than just hue. Diverging palettes or color-blind friendly sequential palettes are recommended. A good option from RColorBrewer (which can be used with `scale_fill_brewer` in ggplot2) for

a sequential scale that is generally color-blind safe is "YlGnBu" (Yellow-Green-Blue) or "Blues". These palettes offer distinct changes in lightness.

6. Explain the difference between sequential and categorical color scales.

- **Sequential Color Scales:** These are used for ordered numerical data (quantitative data) where values progress from low to high. They typically use variations in lightness, saturation, or a gradual change in hue to represent this progression. The primary purpose is to show magnitude and order, making it easy to discern "more" or "less" of a variable. Examples include a gradient from light blue to dark blue for temperature, or light green to dark green for population density.
- **Categorical Color Scales (also known as Qualitative or Diverging for categorical data):** These are used for unordered, discrete categories (qualitative data) where there is no inherent order between groups. They typically use distinct hues to differentiate between categories, ensuring each category is easily distinguishable from others. The primary purpose is to show differences between groups. Examples include using red for 'North', blue for 'South', green for 'East', etc., where the colors themselves don't imply any ordering or magnitude between the regions.

Question 4: Diverging Color Scales for Data Representation

Scenario: You have collected data on student performance, specifically their deviation from the class average in a recent exam. Positive values indicate performance above the average, and negative values indicate performance below the average. You want to visualize this data to quickly identify students who are significantly above or below the average.

Install and load ggplot2 if not already installed

```
if (!requireNamespace("ggplot2", quietly = TRUE)) {
```

```
  install.packages("ggplot2")
```

```
}
```

```
library(ggplot2)
```

1. Create the dataset for Question 4

```
student_deviation <- data.frame(
```

```
  Student = c("Alice", "Bob", "Charlie", "David", "Eve", "Frank", "Grace", "Heidi"),
```

```
  Deviation = c(15, -8, 20, -12, 5, 18, -3, 10)
```

```
)
```

```
print("Dataset for Question 4:")
```

```
print(student_deviation)
```

```
# Order students by deviation for better visualization, useful for showing the range
```

```
student_deviation$Student <- factor(student_deviation$Student, levels =  
student_deviation$Student[order(student_deviation$Deviation)])
```

```
# 2. Write an R program to create a bar chart using a diverging color scale based on  
Deviation.
```

```
bar_chart_diverging_color <- ggplot(student_deviation, aes(x = Student, y = Deviation, fill  
= Deviation)) +
```

```
  geom_bar(stat = "identity") +
```

```
  scale_fill_gradient2(low = "red", mid = "white", high = "steelblue",
```

```
    midpoint = 0, # Important: midpoint at 0 for deviation data
```

```
    name = "Deviation from Average") +
```

```
  labs(  
    title = "Student Deviation from Class Average with Diverging Color Scale",
```

```
    x = "Student",
```

```
    y = "Deviation from Average"
```

```
  ) +
```

```
  theme_minimal()
```

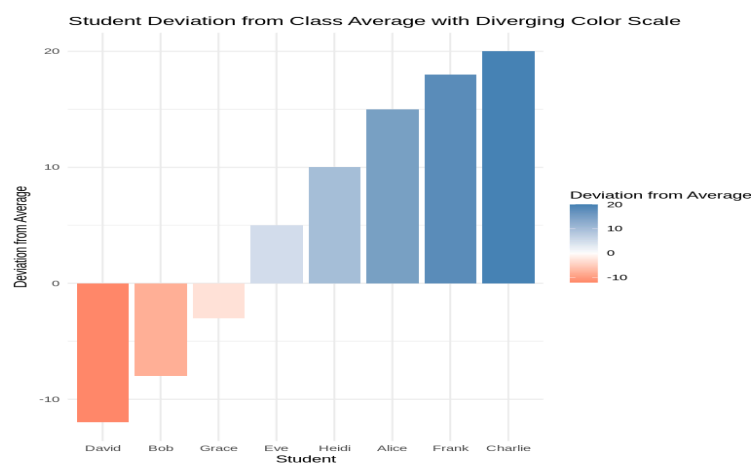
```
print(bar_chart_diverging_color)
```

```
"Dataset for Question 4:"
```

```
Student Deviation
```

```
1 Alice      15
```

2	Bob	-8
3	Charlie	20
4	David	-12
5	Eve	5
6	Frank	18
7	Grace	-3
8	Heidi	10



Interpretation for Question 4

3. Describe the effectiveness of using a diverging color scale for this data.

A diverging color scale is highly effective for visualizing the deviation data. It clearly and intuitively highlights both positive and negative deviations from a central, neutral value (in this case, the class average, represented by 0). Students performing above average are easily distinguished by one color (e.g., blue), while those below average are distinguished by another (e.g., red). The intensity of the color also indicates the magnitude of the deviation, making it easy to spot the highest and lowest performers relative to the average at a glance. The neutral color (white) at the midpoint clearly indicates where the deviation is zero or very close to zero.

4. Explain how you chose the midpoint for the color scale and why it's appropriate.

The midpoint for the color scale was chosen as 0. This is appropriate because the data represents deviations from an average, where 0 signifies exactly the class average (no deviation). By setting midpoint = 0 in `scale_fill_gradient2`, the color scale correctly centers around this meaningful neutral value, visually separating positive and negative values with distinct hues and intensity changes on either side. This allows for a direct visual interpretation of whether a student is above or below the average, and by how much.

5. In what scenarios would a diverging color scale be more appropriate than a sequential or categorical scale?

A diverging color scale is most appropriate in scenarios where:

- **Data has a meaningful central or neutral value:** This central value typically represents a baseline, average, or threshold from which values can diverge in two directions (e.g., above/below average, positive/negative change, profit/loss).
- **Both ends of the spectrum are equally important:** When you want to emphasize and differentiate between extreme low values and extreme high values, with a clear distinction from values near the center.
- **Comparing two opposing concepts:** For instance, showing political leanings (liberal vs. conservative), temperature anomalies (colder vs. warmer than average), or public opinion (approve vs. disapprove).

It is less suitable for data that is strictly ordered from low to high without a natural central pivot (use sequential scale) or for purely distinct, unordered categories (use categorical scale).

Question 5: Faceting for Multi-variable Visualization

Scenario: You have sales data for different product categories across various regions. You want to visualize the sales trend over time for each product category, but also want to easily compare these trends across different regions.

Dataset: Category, Region, Month, Sales (numerical data for 12 months for 3 categories and 2 regions).

Tasks:

1. Create the dataset for Question 5.
2. Write an R program to create a line plot showing sales trends over months, using `facet_wrap` to separate plots by Region and Category.
3. Explain the primary advantage of using faceting in this visualization.
4. Describe how `facet_wrap` differs from `facet_grid` and when you would choose one over the other

```
if (!requireNamespace("ggplot2", quietly = TRUE)) {  
  install.packages("ggplot2")  
}  
library(ggplot2)  
if (!requireNamespace("tidyr", quietly = TRUE)) {  
  install.packages("tidyr")  
}  
library(tidyr)  
if (!requireNamespace("dplyr", quietly = TRUE)) {  
  install.packages("dplyr")  
}
```

```

library(dplyr)
data_q5 <- expand.grid(
  Category = c("Electronics", "Clothing", "Home Goods"),
  Region = c("North", "South"),
  Month = 1:12
) %>%
mutate(
  Sales = round(runif(n(), min = 100, max = 500) +
    ifelse(Category == "Electronics", 100, 0) +
    ifelse(Region == "North", 50, 0) +
    sin(Month/2) * 50, 0)
)

print("Dataset for Question 5 (first 10 rows):")
print(head(data_q5, 10))

line_plot_faceted <- ggplot(data_q5, aes(x = Month, y = Sales, color = Category)) +
  geom_line() +
  geom_point() +
  facet_wrap(~ Region + Category, scales = "free_y", ncol = 2) +
  labs(
    title = "Sales Trend Over Time by Category and Region",
    x = "Month",
    y = "Sales",
    color = "Product Category"
  ) +
  theme_minimal()

print(line_plot_faceted)

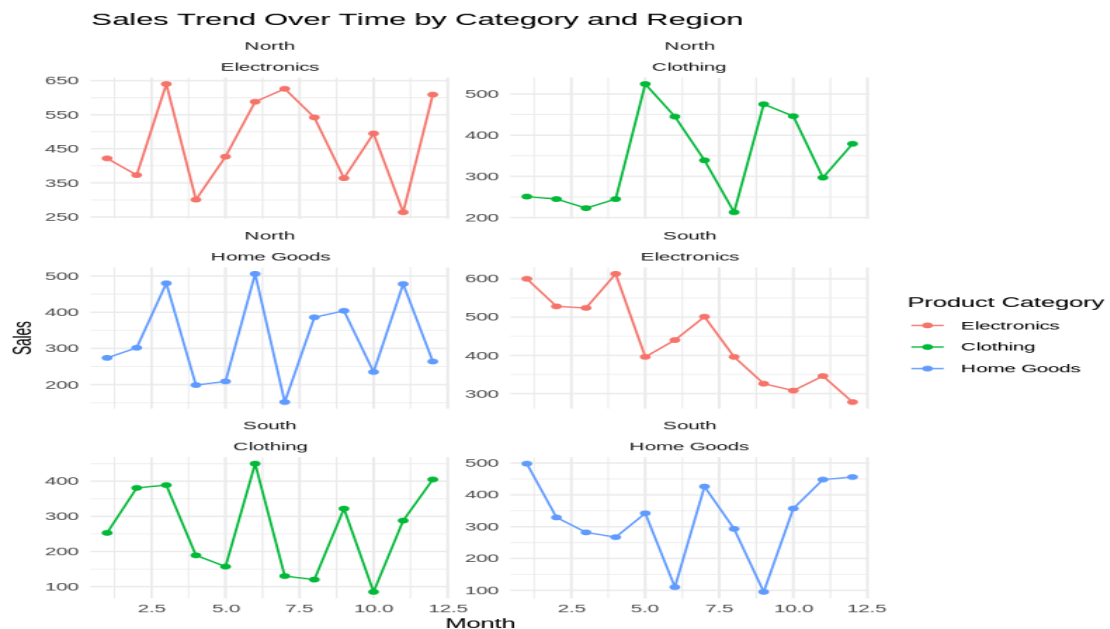
```

Dataset for Question 5 (first 10 rows):"

	Category	Region	Month	Sales
1	Electronics	North	1	422
2	Clothing	North	1	251
3	Home Goods	North	1	274
4	Electronics	South	1	600
5	Clothing	South	1	253
6	Home Goods	South	1	498
7	Electronics	North	2	373
8	Clothing	North	2	245

9 Home Goods North 2 302

10 Electronics South 2 528



Interpretation for Question 5

3. Explain the primary advantage of using faceting in this visualization.

The primary advantage of using faceting (e.g., `facet_wrap` or `facet_grid`) in this visualization is its ability to **disaggregate and compare data across multiple categorical variables without cluttering a single plot**. Instead of trying to cram all categories and regions into one complex plot with many overlapping lines, faceting creates a grid of smaller, individual plots. Each small plot represents a specific combination of Region and Category. This makes it much easier to:

- **Identify trends within each group:** You can clearly see the sales trend over months for "Electronics in North," "Clothing in South," etc.
- **Compare trends across groups:** By placing these plots adjacent to each other, it's straightforward to visually compare how sales trends for "Electronics" differ between the "North" and "South" regions, or how "Electronics" compares to "Clothing" within the same region.

- **Reduce overplotting:** Prevents lines and points from obscuring each other, leading to a clearer and more readable visualization.

4. Describe how `facet_wrap` differs from `facet_grid` and when you would choose one over the other.

Both `facet_wrap` and `facet_grid` are used to create small multiples (grids of plots) based on categorical variables, but they structure these grids differently:

- **`facet_wrap(~ var1 + var2, ...)`:**
 - **Structure:** Wraps a 1D ribbon of panels into 2D. It creates a single row or column of plots, and if there are too many, it wraps them into multiple rows/columns. You specify a single formula (e.g., `~ variable` or `~ var1 + var2`).
 - **Layout:** The number of rows and columns can vary depending on the number of levels in the faceting variable(s) and the `ncol/nrow` argument. It tries to make the layout efficient.
 - **Use Cases:** Best when you have **one or more variables you want to combine to create a set of panels**, and the primary goal is to **display all combinations without strict row/column assignment**. It's useful when the order or direct comparison between specific rows/columns of facets isn't paramount, but rather seeing all individual facets clearly.
- **`facet_grid(row_var ~ col_var, ...)`:**
 - **Structure:** Creates a 2D grid of panels defined explicitly by two faceting variables. The variable before `~` defines the rows, and the variable after `~` defines the columns. You can use `.` if you only want to facet by rows or columns (e.g., `row_var ~ .` or `~ col_var`).
 - **Layout:** The layout is fixed based on the unique levels of the row and column variables. Each cell in the grid represents a unique combination of `row_var` and `col_var`.
 - **Use Cases:** Ideal when you want to **compare specific combinations or look for patterns across the levels of two distinct categorical variables**. It's particularly powerful when you need to maintain a fixed relationship between the rows and columns of your facet grid, facilitating structured comparisons (e.g., comparing all regions for one category vertically, and all categories for one region horizontally).

When to choose:

- Choose `facet_wrap` when you have **one or more variables you want to facet by, and you don't have a strong hierarchical or structured comparison need** between the facet dimensions. It's more flexible with layout.
- Choose `facet_grid` when you have **two specific variables you want to use to define the rows and columns of your plot grid**, and the **structured comparison across those two dimensions is important**.